Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Lagrange's method of undetermined coefficient

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Lagrange's method of undetermined coefficient

1 Lagrange's method of undetermined coefficient

Study focus: Lagrange's method of undetermined coefficient. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000234 · EV0000235

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In the previous session we discussed about the extrema of a function of two independent variables, right?

Key details

- And we did solve some problems also.
- Now we are going to study another method which is called the Lagrangius method of undetermined multipliers.
- So such problems wherein we get a val the function along with the constraint can be solved by using Lagrange's method of undetermined multiplier.
- The next thing that is you need to do is you solve the first order partial derivatives.
- This is an advantage.
- You remember what we did here?
- We had x square plus y square plus z square equal to 1.
- So this z square we substituted back here and then eliminated that z that can be avoided if we use this method.
- That's why it's called an multiplier because this is like a scaling of the constraint, right?
- And it is undetermined, so we don't need to worry about what is the value of that lambda, but we'll calculate the value, okay?
- So calculate the first order partial derivatives, equate it to zero and use the constraint.
- Constraint is phi of x y z.
- for to solve lambda x y and z.
- Three of them I got here f x 0, f y 0 and f z 0.
- I'll I'll use that constraint and then I'll s use that to solve the lambda value.
- The stationary points are nothing but the critical points, right?

- You remember what we did in the previous case?
- Ah and second advantage is this method can be extended to n parameters, right?
- If I have x of x y z w, then you write the equation by introducing lambda, that's all you have to do.
- But the disadvantage is you will not be able to know what is the nature of the stationary point or what is the nature of the critical point.
- That means you will not be able to decide whether that will represent a maxima point or a minimum point, that is the disadvantage.
- All you can do is get the stationary points.
- that it will represent either maximum or minimum.
- But I cannot verify if I have to verify this R T minus S square to be greater than 0 and R is less than 0 or R is greater than 0, I have to go back to the previous method only.
- That's the disadvantage of this.
- But definitely to find the critical point you can make use of this.
- So that means that might look something like let me try to sketch that.
- I don't know how well I can sketch.

Example or application from the lecture

- So, in the first problem if you recollect what we did is we studied the temperature if the temperature function was XYZ was 400 XYZ square, this was given to us, and then we had to calculate what is the maximum temperature on the surface of a sphere.
- from the given function.
- Let's try a problem, maybe things will be little more clear, right?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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Core explanation · 05:00–09:59

This so I have the sphere.

Key details

- And then now what I want is I have to inscribe a rectangle in it.
- If this has to have a maximum value, then it has to be a cube.
- It's x square plus y square plus z square equal to r square with the center of course at the origin and radius r.
- And what is the volume of the rectangular solid, F of X Y Z?
- Two eight x.
- That's because I have taken two x and then two y and two z.
- Which one I need to maximize or minimize, I need to worry about this.
- What is the constraint inside the sphere.
- The volume is like length into breadth into height, so you can take that as x y z.
- And what is FY?
- F y will be x z plus lambda into f y so I have to I'll get two y.
- Plus lambda into two set.

- I can take out z common and then I get lambda y plus two lambda plus y equal to zero.
- Don't take this 8 also it is absolutely fine.
- But what you can do is if you multiply this first equation with x, I will get x y z plus 2 x square lambda is 0.
- I will do the same thing for the next equation.
- I will take x y z plus to y square lambda equal to 0.
- What do I get if I solve these two equations?
- I will subtract them.
- I will get 2x square lambda minus 2y square lambda equal to 0.
- This will give me lambda equal to 0 or x is equal to plus or minus 1, right.
- Because it's a measure minus y does not matter for me, so I will get x is equal to y.
- We never use the fact that I have to take R T minus S square and then find out R greater than zero or less than zero.
- In such problems the Slagrange's method of undetermined multipliers are very helpful.
- The question is find the points closest to the origin of the hyperbolic cylinder.
- X square plus Y square plus Z square because a sphere can be taken as a point.
- Maybe if I take this as R instead of that.
- As R becomes R square maybe.

Example or application from the lecture

- Let's prove that mathematically, okay.
- That was not necessary because the question said that maximum volume, they have already given that it is maximum.
- Let's try another problem.

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Core explanation · 10:00–14:59

If R becomes zero, I will get the s smallest a point which is nothing but the origin.

Key details

- On the hyperbolic cylinder, x square minus z square equal to.
- Or if you want to call g of x y z, g of x y z is x square minus z square minus 1.
- I need to find XY Z and that lambda.
- First let me calculate what's f x is.
- F x will be two x plus two x lambda, right?
- and f y will be again two y plus a zero, there is no y term there, and f z will be two z minus two lambda z.
- But I have another condition right which is x square minus z square minus one equal to zero.
- I have to equate them to zero, right?
- That's that's what is the condition, f x is zero, f y is zero and f z is zero.
- What about the first equation if you observe carefully the first equation is 2x plus 2x lambda is 0.

- This gives me x equal to 0 or lambda is equal to minus 1.
- But can x be zero?
- There is a problem with that.
- If x becomes zero, if x is zero, then what do I get?
- I'll get z square to be minus one.
- So that means my z will not be in the real plane at all.
- That's a problem I have, right?
- So that means x is equal to zero is not possible.
- But I can take definitely lambda equal to minus one.
- Now what is the next equation I have?
- I have two z.
- Ah 2 z minus 2 lambda z equal to 0.
- This will give me z equal to 0 or lambda equal to 1.
- That is fine for me.
- But if z is 0 what is the value of x that I get?
- I have x square minus z square minus 1 is 0.
- That means the point that will be very close to the origin will be plus or minus 1, 0 and 0.
- Is it clear?
- So you remember if you are using Lagrangius method of multiplier, the condition will also be mentioned.
- That means that point is minimum only for you.
- Here it is closest.
- What is the rectangular solid of maximum volume that can be inscribed in a sphere.
- So whenever you have a constraint then you can definitely make use of Lagrange's method of undetermined multiplier, that's one thing.
- Similarly the first problem that we did in the previous session can also be solved using Lagrangian's method of undetermined multipliers.
- You practice some more problems then concept should be more clear to you, right?

Example or application from the lecture

- My axillary equation would look like f of x y z is equal to this is the given function x I want to make this as small as possible with the given condition that x square minus z square minus.
- In the previous example it was maximum right.
- The second thing is whenever they have already given the condition that you have to maximize the temperature.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Lagrange's method of undetermined coefficient transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00062	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00063	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...
SRC00064	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare...

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C Revision and retrieval practice